$SL(n,\mathbb{C}) = \{ A \in GL(n,\mathbb{C}) \mid \det(A) = 1 \}$

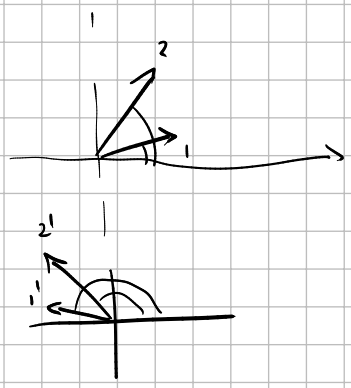
if $\det : GL(n,\mathbb{C}) \to \mathbb{C}$ $\qquad SL(n,\mathbb{C}) = \det^{-1}(\{1\})$

↳ continuous → pre image of closed set is closed

$GL(2,\mathbb{R})$ set of invertible linear transformations of $\mathbb{R}^2$

if $\det(A) < 0$ → changes orientation

reflections have det -1



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if $|\det(A)| = 1$ → vectors keep areas

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why orientation important? $\mathbb{1}$ has det 1

an infinitesimal transformation must be connected to $\mathbb{1}$

G is MLG. Connected component of G (G_0) is the set of all points in G that are path-connected to identity

$g \in G_0$ if there is a curve $\gamma : [a,b] \to G$ s.t. $\gamma(a) = e$, $\gamma(b) = g$

→ if $g \in G_0$ then we can write g from Lie algebra

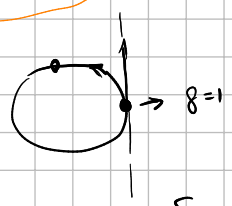
Specifically, there are $X_1, \dots X_s$ s finite in Lie algebra s.t.

$g = e^{X_1} e^{X_2} \dots e^{X_s}$ $\qquad e^{X_1} e^{X_2} \neq e^{X_2} e^{X_1}$

---

suppose $\boxed{g = e^{tX}}$ $\quad X \in \mathfrak{g}$

$= \mathbb{1} + tX + o(t^2)$



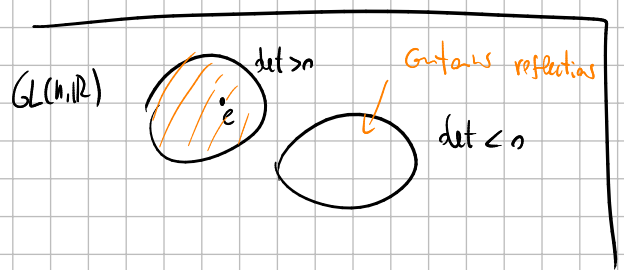
$U(1) = \{ z \in \mathbb{C} \mid |z| = 1 \}$

$= \{ e^{i\theta} \mid \theta \in \mathbb{R} \}$

$\mathfrak{u}(1) = \mathrm{span}_{\mathbb{R}} \{ i \} = \{ ix \mid x \in \mathbb{R} \}$

$z \in U(1)$

$z = e^{tX}$ $\qquad X = i$



Exponential of matrix $\quad X \in M_n(\mathbb{C}) \qquad e^X$

$$\begin{tikzcd} G \arrow[d, "\rho"'] & \mathfrak{g} \arrow[l, "\exp"'] \arrow[d, "\rho_*"] \\ \mathrm{Aut}(V) & \mathrm{End}(V) \arrow[l, "\exp"] \end{tikzcd}$$

difficult $\qquad$ easy

Idea: find reps of $\mathfrak{g}$ and exponentiate them. But not all of them end up as reps of G.

For $G = SO(3)$ the ones that fail are the spinor representations

All works is G simply connected

(ex: $SU(2)$ simply connected)

$SU(2) = \widetilde{SO(3)}$ (simply connected cover of $SO(3)$)

double cover of $SO(n)$ is $Spin(n)$

accidental isomorphism

$SL(2,\mathbb{R}) \cong Sp(2,\mathbb{R})$

$$SL(n,\mathbb{C}) = \{A \in GL(n,\mathbb{C}) \mid \det(A) = 1\}$$

$$\underline{\mathfrak{sl}}(n,\mathbb{C}) = \{X \in M_n(\mathbb{C}) \mid e^{tX} \in SL(n,\mathbb{C}) \ \forall t \in \mathbb{R}\}$$

$$= \{X \in M_n(\mathbb{C}) \mid \det(e^{tX}) = 1 \ \forall t \in \mathbb{R}\}$$

① $\det(e^{tX}) = 1 \ \forall t \iff e^{t\,\mathrm{tr}(X)} = 1 \ \forall t$

$\Rightarrow$ $e^{t\,\mathrm{tr}(X)} = 1 \ \forall t$ then $\underbrace{\left.\frac{d}{dt}\right|_{t=0} e^{t\,\mathrm{tr}(X)}}_{= \mathrm{tr}(X)} = \underbrace{\left.\frac{d}{dt}\right|_{t=0} 1}_{=0} \iff \mathrm{tr}(X) = 0$

$\Leftarrow$ if $\mathrm{tr}(X) = 0$ then $e^{t\,\mathrm{tr}\,X} = e^0 = 1 \ \forall t$

$$\underline{\mathfrak{sl}}(n,\mathbb{C}) = \{X \in M_n(\mathbb{C}) \mid \mathrm{tr}(X) = 0\}$$

ex $\quad \underline{\mathfrak{sl}}(n,2) = \{X \in M_2(\mathbb{C}) \mid \mathrm{tr}(X) = 0\} = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(\mathbb{C}) \,\middle|\, a + d = 0 \right\}$

$\underline{sl}(n,2) = \left\{ \begin{pmatrix} a & b \\ c & -a \end{pmatrix} \middle| \; a,b,c \in \mathbb{C} \right\}$ needs to be a real vector space

$= \left\{ \begin{pmatrix} x+iy & u+iv \\ s+it & -x-iy \end{pmatrix} \middle| \; x,y,u,v,s,t \in \mathbb{R} \right\}$

basis : $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & i \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ i & 0 \end{pmatrix}$

because $[\cdot,\cdot]$ is bilinear, we just need to know how it acts on a basis

basis elements are called <u>generators of G</u>

$\underline{so}(n) = \left\{ X \in M_n(\mathbb{R}) \middle| \; e^{tX} \in SO(n) \;\; \forall t \right\}$

- $\det(e^{tX}) = 1 \;\; \forall t \iff tr(X) = 0$
- $(e^{tX})^T = (e^{tX})^{-1} \;\; \forall t \iff e^{tX^T} = e^{-tX} \;\; \forall t \iff X^T = -X$

$\underline{so}(n) = \left\{ X \in M_n(\mathbb{R}) \middle| \; X^t = -X \right\}$

$\underline{so}(2) = \left\{ X \in M_2(\mathbb{R}) \middle| \; X^t = -X \right\} = \left\{ \begin{pmatrix} 0 & x \\ -x & 0 \end{pmatrix} \middle| \; x \in \mathbb{R} \right\} = \mathrm{span}_{\mathbb{R}} \left\{ \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \right\}$

$\exp\begin{pmatrix} 0 & x \\ -x & 0 \end{pmatrix} = \begin{pmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{pmatrix} = R(x)$ rotation matrix with angle x

$X_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ generator of 2D rotations

$R(\theta) = e^{\theta X_0}$

<u>example of rep.</u> $V = \mathrm{span}_{\mathbb{C}} \{ |\psi\rangle \}$

$\rho(R(\theta)) |\psi\rangle = e^{i\theta} |\psi\rangle$